STATISTICS 652

LECTURE 10 - PART 2 - 2026

- MULTIVARIATE NORMAL - MARGINAL DISTRIBUTIONS
  - SEE COMPLETION OF LECTURE 10

- CONDITIONAL DISTRIBUTIONS: ASSUME $\Sigma_{22} > 0$. FIND $\underset{\sim}{X}_1 | \underset{\sim}{X}_2$)

  i.e. CONDITIONAL DISTRIBUTION OF $\underset{\sim}{X}_1$ GIVEN $\underset{\sim}{X}_2$.

LET $\hat{\underset{\sim}{X}}_1 = \underset{\sim}{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\underset{\sim}{X}_2 - \underset{\sim}{\mu}_2)$ and $\underset{\sim}{X}_{1.2} = \underset{\sim}{X}_1 - \hat{\underset{\sim}{X}}_1$

INSIGHT: MULTIVARIATE LINEAR REGRESSION OF $\underset{\sim}{X}_1$ ONTO $\underset{\sim}{X}_2$ $\leftrightarrow$ $\hat{\underset{\sim}{X}}_1$; RESIDUAL $\underset{\sim}{X}_{1.2}$

SO $\begin{pmatrix} \underset{\sim}{X}_{1.2} \\ \underset{\sim}{X}_2 \end{pmatrix} = \begin{bmatrix} I & -\Sigma_{12}\Sigma_{22}^{-1} \\ 0 & I \end{bmatrix} \begin{pmatrix} \underset{\sim}{X}_1 \\ \underset{\sim}{X}_2 \end{pmatrix} + \begin{pmatrix} \underset{\sim}{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}\underset{\sim}{\mu}_2 \\ \underset{\sim}{0} \end{pmatrix}$

$\sim N_q\left(\begin{pmatrix} 0 \\ \underset{\sim}{\mu}_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11.2} & 0 \\ 0 & \Sigma_{22} \end{pmatrix} \right)$

WHERE $\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$ (RESIDUAL VAR-COV MATRIX)

$\Rightarrow \underset{\sim}{X}_{1.2}$ and $\underset{\sim}{X}_2$ are independent (LEMMA 10.7)

$\Rightarrow \underset{\sim}{X}_1 | \underset{\sim}{X}_2 = (\hat{\underset{\sim}{X}}_1 + \underset{\sim}{X}_{1.2}) | \underset{\sim}{X}_2 \sim \hat{\underset{\sim}{X}}_1 | \underset{\sim}{X}_2 + \underset{\sim}{X}_{1.2}$

with $\hat{\underset{\sim}{X}}_1$ being CONSTANT GIVEN $\underset{\sim}{X}_2$

$\Rightarrow \underset{\sim}{X}_1 | \underset{\sim}{X}_2 \sim N_{q_1}(\hat{\underset{\sim}{X}}_1, \Sigma_{11.2})$ (NO CALCULUS NEEDED)

NOTE: i) $E[\underset{\sim}{X}_1 | \underset{\sim}{X}_2] = \underset{\sim}{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\underset{\sim}{X}_2 - \underset{\sim}{\mu}_2)$ i.e. REGRESSION IS LINEAR

ii) $\text{VAR}[\underset{\sim}{X}_1 | \underset{\sim}{X}_2] = \Sigma_{11.2}$ HOMOSEDASTIC, i.e. $\Sigma_{11.2}$ DOES NOT DEPEND ON $\underset{\sim}{X}_2$.

o OTHER PROPERTIES OF MULTIVARIATE NORMAL

MOMENT GENERATING FUNCTION

i) $\underset{\sim}{Z} \sim N_q(\underset{\sim}{0}, I) \Rightarrow M_{\underset{\sim}{Z}}(\underset{\sim}{t}) = E[e^{\underset{\sim}{t}'\underset{\sim}{Z}}] = \prod_{j=1}^{q} E[e^{t_j Z_j}]$

$= \prod_{j=1}^{q} M_Z(t_j) = \prod_{j=1}^{q} e^{\frac{1}{2}t_j^2} = e^{\frac{1}{2}\sum_{j=1}^{q} t_j^2} = e^{\frac{1}{2}\underset{\sim}{t}'\underset{\sim}{t}}$.

ii) $\underset{\sim}{X} \sim N_q(\underset{\sim}{\mu}, \Sigma), \Sigma > 0 \Rightarrow \underset{\sim}{X} \sim \Sigma^{\frac{1}{2}}\underset{\sim}{Z} + \underset{\sim}{\mu} \Rightarrow$

$M_{\underset{\sim}{X}}(\underset{\sim}{t}) \equiv E[e^{\underset{\sim}{t}'\underset{\sim}{X}}] = E[e^{\underset{\sim}{t}'\Sigma^{\frac{1}{2}}\underset{\sim}{Z} + \underset{\sim}{t}'\underset{\sim}{\mu}}] = e^{\underset{\sim}{t}'\underset{\sim}{\mu}} M_Z(\Sigma^{\frac{1}{2}}\underset{\sim}{t})$

$= e^{\underset{\sim}{t}'\underset{\sim}{\mu} + \frac{1}{2}\underset{\sim}{t}'\Sigma\underset{\sim}{t}}$

CHARACTERISTIC FUNCTION:

$\phi_X(\underset{\sim}{t}) = E[e^{i\underset{\sim}{t}'\underset{\sim}{X}}] = e^{i\underset{\sim}{t}'\underset{\sim}{\mu} - \frac{1}{2}\underset{\sim}{t}'\Sigma\underset{\sim}{t}}$

NOTE: ONE COULD DEFINED THE MULTIVARIATE NORMAL BY ITS CHARACTERISTIC FUNCTION, HOWEVER, IN DOING SO, ONE MUST VERIFY IT IS A VALID CHACTERISTIC FUNCTION.

RECALL FROM LECTURE 6 - PAGE 6

RANDOM SAMPLE: $\underset{\sim}{X}_1, \ldots, \underset{\sim}{X}_n$ iid $N_q(\underset{\sim}{\mu}, \Sigma)$, $\Sigma > 0$

- FULL RANK $k = q + \frac{1}{2}q(q+1)$ EXPONENTIAL FAMILY
- NATURAL SUFFICIENT STATISTICS

$$\left\{ T_1 = \sum_{i=1}^n \underset{\sim}{X}_i \underset{\sim}{X}_i' \,,\; T_2 = \sum_{i=1}^n \underset{\sim}{X}_i \right\}$$

- LIKELIHOOD EQUATIONS: $T_1 = E(T_1) = n(\Sigma + \underset{\sim}{\mu}\underset{\sim}{\mu}')$

$$T_2 = E[T_2] = n\underset{\sim}{\mu}$$

- MLE: $\hat{\underset{\sim}{\mu}} = \bar{\underset{\sim}{X}}$ and $\hat{\Sigma} = S_n$ (EMPIRICAL COVARIANCE MATRIX)

PROVIDED S_n IS NON-SINGULAR.

- FOR S_n SINGULAR (WHICH OCCURS IF $n \leq q$), the MLE DOES NOT EXIST SINCE LIKELIHOOD IS NO BOUNDED ABOVE

TO VERIFY: SHOW log likelihood can be expressed as:

$$\ell(\underset{\sim}{\mu}, \Sigma) = -\frac{n}{2}\left\{ TR(\Sigma^{-1}S_n) + \log\det\Sigma + (\bar{\underset{\sim}{X}} - \underset{\sim}{\mu})'\Sigma^{-1}(\bar{\underset{\sim}{X}} - \underset{\sim}{\mu}) \right\}$$

then consider $\underset{\sim}{\mu} = \bar{\underset{\sim}{X}}$ and $\Sigma_\lambda = (I + \lambda \underset{\sim}{b}\underset{\sim}{b}')^{-1}$ $\lambda > 0$

and where $S_n \underset{\sim}{b} = 0$, $\|\underset{\sim}{b}\| = 1$

$$\Rightarrow \ell(\bar{\underset{\sim}{X}}, \Sigma_\lambda) = -\frac{n}{2} TR(S_n) + \log\det(I + \lambda \underset{\sim}{b}\underset{\sim}{b}')$$

$$= -\frac{n}{2} TR(S_n) + (1+\lambda) \to \infty \text{ as } \lambda \to \infty. \quad \square$$

ALSO RECALL BY LEHMANN-SCHEFFÉ

a) CONSIDER ANY $\hat{\underset{\sim}{\mu}}$ such that $E(\hat{\underset{\sim}{\mu}}) = \underset{\sim}{\mu}$. THEN SINCE $E(\bar{\underset{\sim}{X}}) = \underset{\sim}{\mu}$,

$$\text{VAR}(\underset{\sim}{a}'\bar{\underset{\sim}{X}}) = \underset{\sim}{a}'\text{VAR}(\bar{\underset{\sim}{X}})\underset{\sim}{a} \leq \text{VAR}(\underset{\sim}{a}'\hat{\underset{\sim}{\mu}}) = \underset{\sim}{a}'\text{VAR}(\hat{\underset{\sim}{\mu}})\underset{\sim}{a}, \quad \underset{\sim}{a} \in \mathbb{R}^q$$

b) CONSIDER ANY $\hat{\Sigma}$ such that $E(\hat{\Sigma}) = \Sigma$. THEN SINCE $E(S) = \Sigma$,

$$\text{VAR}(\underset{\sim}{a}' S \underset{\sim}{b}) \leq \text{VAR}(\underset{\sim}{a}' \hat{\Sigma} \underset{\sim}{b}) \quad \text{for } \underset{\sim}{a} \in \mathbb{R}^q \text{ and } \underset{\sim}{b} \in \mathbb{R}^q$$

$$\text{ALSO } \text{VAR}(\text{TR}(AS)) \leq \text{VAR}(\text{TR}(A\hat{\Sigma})) \quad \text{for } A_{q\times q}$$

DEF 10.8. LOEWNER PARTIAL ORDERING OF SYMMETRIC MATRICES OF ORDER q.

$$A_1 \leq A_2 \Leftrightarrow A_2 - A_1 \geq 0, \quad A_1 < A_2 \Leftrightarrow A_2 - A_1 > 0$$

$$A_1 \geq A_2 \Leftrightarrow A_1 - A_2 \geq 0, \quad A_1 > A_2 \Leftrightarrow A_1 - A_2 > 0.$$

NOTE: $A_1 \geq A_2$ AND $A_1 \neq A_2 \not\Rightarrow A_1 > A_2$ (HW)

So, a) $\text{VAR}(\bar{\underset{\sim}{X}}) \leq \text{VAR}(\hat{\underset{\sim}{\mu}})$ $\quad (\text{VAR}(\bar{\underset{\sim}{X}}) = \frac{1}{n}\Sigma)$

b) $\text{VAR}(S) \leq \text{VAR}(\hat{\Sigma})$ EVEN FOR $n \leq q$ (SINGULAR S)

WHERE $\text{VAR}(S) \equiv \text{VAR}(\text{VEC}(S))$ with $\text{VEC}(A_{q\times p}) = \begin{pmatrix} \underset{\sim}{a}_1 \\ \vdots \\ \underset{\sim}{a}_p \end{pmatrix}$

NOTE: $\text{TR}(AS) = \text{VEC}(A)'\text{VEC}(S)$